

Assignment 8: Builder Journal

Trace — Personal Productivity and Task Execution Agent

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What I Built

I built Trace, a personal productivity agent that helps me manage tasks and notes through natural conversation. It can create, update, and complete tasks, search and save notes, extract action items from meeting notes, generate a work plan for the day, and draft reminders and follow-up emails. Before it makes any change that writes data or can't be undone, it shows me an approval card so I can approve, reject, or edit the action first, and it keeps a log of every run so I can see exactly what it did.

The Most Difficult Technical Problem

The most difficult problem I ran into was with the API. Everything had been working fine, and then suddenly I stopped getting any response back from it at all. My first guess was that I had hit a rate limit, so I created a second account to work around it. That didn't fix the problem, though — my tasks still weren't being created, and I was left without a clear answer for what was actually going wrong.

How I Solved It

As of writing this, I haven't fully solved it yet. Creating a second account ruled out a simple per-account rate limit, which tells me the real cause is something else — possibly an invalid or expired key, a quota/billing issue on the provider's side, or a bug in how my app is calling the API. My next step is to actually check the error response (status code and message) instead of assuming it's a limit, since that will tell me which of these it really is.

Tool-Calling Errors I Observed

I didn't track these systematically during this phase. The clearest one I noticed was that task-creation requests were failing silently while the API issue was happening — the `create_task` call was not returning a usable response, so no task was actually created.

Agent Behavior That Surprised Me

What surprised me most was how strictly the agent stuck to asking for approval before any write action, even on small changes I expected it to just handle on its own, like updating a due date. I had assumed it would only pause for the “big” actions like deleting or completing a task, but it treated every write the same way. It also surprised me how much the API outage exposed — the agent didn't crash or give a confusing error, it just quietly stopped producing results, which made the problem harder to notice at first than I expected.

What Failed During Testing

API testing failed during this phase — while trying to verify that task creation worked end-to-end, the API stopped responding, which meant I couldn't confirm whether the `create_task` and `update_task` tools were actually behaving correctly.

What I Would Redesign

If I redesigned this now, I would add a clearer failure state for when the API stops responding, instead of the request just silently going nowhere. A visible “could not reach the model provider, try again” message would have saved me a lot of the confusion I had figuring out whether it was my code, my account, or the provider itself. I'd also add basic logging around every outgoing API call so I can see the actual response instead of guessing.

What I Learned About Agent Reliability

This experience taught me that an agent is only as reliable as the weakest link underneath it, and that link is often outside my own code entirely. I had assumed most reliability problems would come from my own logic — wrong tool picked, bad argument, missed edge case — but a silent API failure showed me that the connection to the model provider itself needs the same level of error handling and visibility as everything else in the system.

Goals for Week 4

My main goal for Week 4 is to properly resolve the API issue — by checking the actual error responses instead of guessing — so task creation is reliable again. After that, I want to run the full evaluation test set for real against the live app and record genuine results, and start addressing the highest-priority items from my security review, starting with API key handling since that's directly connected to the problem I ran into this week.